

Di Li

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INDUSTRY EXPERIENCE

Control Engineer

04/2025–Present

Heraeus Quartz North America

- Design and develop new control systems for SZUG (fiber quartz glass manufacturing machines), increasing overall equipment effectiveness.
- Architect software and hardware integration for logic control, tower and puller motion, and temperature/oven power regulation.
- Integrate precision measuring components and develop comprehensive HMIs for system monitoring and operation.
- Pioneer a new oven heating control system and a novel nitrogen pressure control system, boosting product yield.
- Complete electrical schematics and layouts using EPLAN.

Senior Engineering Technician

03/2024–03/2025

IDEXX

- Maintain the manufacturing production line for biomedical sensor products.
- Troubleshoot production equipment and modules, including high speed linear motor transportation system, vision-guided robot pick-and-place, smart vision for quality inspections.
- Optimize PLC coding to decrease machine downtime significantly.
- Innovate the illumination strategy for AOI system, decreased false acceptance rate by 23%.
- Increase the vision guided robot pick-and-place success rate by 15% using improved calibration.

CTO

11/2016–11/2020

Shenzhen Qizhong Intelligent Company

- Lead the development of machine controller hardware and software platforms, including motion control, PLC, and EtherCAT engineering and runtime environments.
- Innovate multiple DSP/FPGA-based embedded controller platforms and software-based open controller platforms.
- Pioneer machine vision algorithm development for industrial environments, specializing in dimension measurement and surface defect detection.
- Direct application development, integrating motion, logic, and vision software to meet diverse performance requirements.
- Develop CAD/CAM interpreters, path planning and optimization algorithms, and HMI designs.
- Provide technical guidance to machine builders in sectors such as consumer electronics, EV battery manufacturing, and solar panel production, supporting applications including dispensing machines, inspection systems, and high-speed pick-and-place robotic modules.

CTO

01/2010–05/2016

Shenzhen Shengyu Intelligent Company

- Lead control and vision system development for the Robotic Stiffener Bonder, an automated vision-guided bonding machine for consumer electronics.
- Direct control and vision development for PCB automated micro-driller manufacturing equipment, including grinding machines, AOI systems, CNC tools, and packaging machines.
- Deliver innovative, cost-effective designs that enabled the company to become a supplier to the world's largest micro-tools manufacturer.
- Spearhead control and vision development for an automatic FPC gold fingers inspection machine (AOI) featuring dimension and surface defect detection.
- Develop the first inspection machine of its kind for the FPC industry, reducing customer product complaint rates by over 50%.

ACADEMIC EXPERIENCE

Full Tenured Professor

11/2001–11/2022

School of Mechanical and Automotive Engineering, South China University of Technology

- **Pioneer machine controller hardware/software co-design** using Model-Based System Engineering including simulation, verification, and automated code generation for controller design and implementation.
- **Innovate multiple hardware/software machine controller platforms** that were successfully commercialized in the China controller market.
- **Design Industry 4.0 Cyber-Physical Production System (CPPS) architectures**, establishing one of the earliest plug-and-produce CPPS environments for customized product manufacturing.
- **Develop AI-driven solutions for flexible job shop dynamic scheduling**, optimizing production efficiency and resource utilization.
- **Implement big-data analytics for machine fault detection**, enabling predictive maintenance and reducing downtime.
- **Pioneer computing and communication co-design for Industrial IoT (IIoT)**, ensuring reliable and real-time data exchange in manufacturing environments.
- **Innovate high-precision machine vision solutions for manufacturing**, including dimensional measurement, AI-driven targeting, and surface defect detection for industrial products.
- **Commercialize China's first 2.5D AOI measurement system in 2005**, a technology still widely applied in the industry today.
- **Developed innovative automated inspection machines**, including systems for Flexible Printed Circuits (FPC) and micro-tool AOI machines, significantly improving quality control capabilities.

EDUCATION

Ph.D. , Control Theory and Application <i>South China University of Technology</i>	06/1993
M.S. , Mechanical Engineering <i>Nanjing University of Aeronautics and Astronautics</i>	04/1988
B.S. , Mechanical Engineering <i>Nanjing University of Aeronautics and Astronautics</i>	06/1985

HONORS AND CONTRIBUTIONS

- Received 8 prestigious innovation awards from industry and academia.
- Member of top-level expert group in China that provides consultancy services for government and industry.
- 50+ patents.
- ~150 papers in smart manufacturing, including highly cited papers. <https://www.researchgate.net/profile/Di-Li-15>
- Most productive author in industry 4.0 based on third-party evaluations.